

Weight–Clock Bifurcation in Cohomological Hénon Moments

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14 August 2026

Abstract

We compare the cohomological structures of the second, third, and fourth chronological moments of a finite-field Fourier–cubic quantization of an area-preserving Hénon map. At every good split prime, projective radial cancellation expresses the normalized n -th moment as minus twice the sum of a pure weight-zero trace E_n and a pure weight-one trace O_n . For a smooth source member their ranks are $(4^n + 5)/3$ and $2(4^n - 4)/3$, hence the total rank is $4^n - 1$. The unchanged Galois normalization forces an exact leading extraction

$$\exp(-\ell_n(s)/n) = \exp\left(\frac{2}{n} \operatorname{Log}_0 L_K^{(S)}(E_n \oplus O_n, ns + 1)\right) H_{n,S}(s),$$

where S is the frozen finite bad set and $H_{n,S}$ is holomorphic and nonzero for $\operatorname{Re} s > 0$. The prime denominator fixes the full clock $u = ns + j$, $j \geq 1$. Its leading weight-one factors share center $s = 0$, whereas the weight-zero centers are $-1/4, -1/6, -1/8$; higher denominator terms also split the odd rail. This mismatch is invariant under consistent Tate relabeling. Fractional leading powers at $n = 3, 4$ cannot arise from a semisimple direct source-native finite-rank K -compatible system retaining the two-weight decomposition and unchanged split-prime trace, although restriction of scalars and Galois counterpackets are not excluded. Clearing denominators leaves a formal odd expected-completion skeleton with exponents 6, 4, 3. Finally, an exact χ_y calculation for the $n = 4$ fivefold identifies a rank-two extreme Hodge piece as the next algebraic-projector gate. No $n = 3, 4$ Hasse–Weil functional equation or full Hénon completion is claimed.

1 Introduction

A dynamical-zeta candidate can possess an exact Euler germ and an operator determinant without yet possessing the analytic structure relevant to a Hilbert–Pólya program. The missing data include a source-native archimedean completion, a functional equation, and a coherent reflection center. Classical Selberg and Ruelle zeta functions acquire such data from trace formulas or geometric spectral identities [17, 15]. The present arithmetic Hénon construction does not have an analogous global trace formula.

The preceding Hénon sequence retained a genuine chronological Fourier–cubic kernel, descended it through a normalized Galois trace, and obtained a normalized-semifinite regularized determinant. Its second, third, and fourth logarithmic moments were then identified with a genus-four curve, a Fano threefold packet, and a fivefold packet, respectively. Those identifications moved the normalized Euler object successively to $\operatorname{Re} s > 1/3$, $\operatorname{Re} s > 1/4$, and $\operatorname{Re} s > 1/5$. The second moment was additionally resummed into modular elliptic factors. This paper asks the next large question: *do the three cohomological moments admit one standard completion without changing their prime clock?*

The answer has a positive and a negative part. The positive structure is a uniform pair of pure-weight packets. If S_n is the Fermat cubic and X_n is the source-ordered $(2, 3)$ complete intersection, then

$$E_n = \mathbf{Q}_\ell(0) \oplus H_{\text{prim}}^{2n-2}(S_n)(n-1), \quad O_n = H^{2n-3}(X_n)(n-2)$$

have weights zero and one, and the exact moment trace is their sum. Their total rank is $4^n - 1$. This identity is a smooth-family theorem; the Hénon source is used only for $n = 2, 3, 4$, where smoothness away from finite bad sets is inherited.

The negative structure comes from the clock rather than from a failure of purity. Field-degree normalization contributes

$$\frac{1}{p-1} = \sum_{j \geq 1} p^{-j},$$

so a standard factor is evaluated at $u = ns + j$. A pure weight- w reflection consequently has center

$$s_{n,j}(w) = \frac{(w+1)/2 - j}{n}.$$

The leading odd factors align at zero, but the leading even factors do not, and later denominator terms split the odd factors. The exact second-moment factorization already proves this bifurcation with factors whose functional equations are known. Thus the obstruction does not rely on conjectural analytic continuation in the higher moments.

Three qualifications are essential. First, center zero is not itself incompatible with an RH normalization: recentering the completed Riemann ξ -function produces a reflection about zero [13]. Our obstruction is the *mismatch* of factorwise centers under one frozen clock. Second, the powers $2/3$ and $1/2$ are canonical Log_0 -germs in a nonvanishing Euler domain, not meromorphic roots. Third, the finite-rank no-go concerns a direct source-native K -compatible packet preserving the same split-prime trace. It does not classify restriction of scalars, Galois-orbit counterpackets, or the normalized-semifinite determinant already constructed.

The standard ingredients of the proof are classical. We use Deligne purity [4], weak Lefschetz [10], the hypersurface Jacobian-ring description [9], and Hirzebruch–Riemann–Roch [12]. Serre and Deligne provide the motivic local-factor, Tate-shift, and expected-completion conventions [18, 5]. Buzzard–Gee and the automorphic normalization literature explain why half-sum shifts are legitimate in a different normalization category [2, 16]; they do not supply an ordinary half Tate motive preserving our coefficients. Regularized-product approaches to Gamma and local factors [14, 6, 7] are close conceptual neighbors, but no full Hénon archimedean determinant is constructed here.

Our new contribution is the source-locked synthesis of these ingredients: the exact two-weight trace and rank identity, the forced exponent $2/n$, the complete center tower, a scoped direct-compatible-system obstruction, and an explicit Hodge projector gate. Standard product stability in motivic λ -rings [11] explains why denominator-cleared products are natural, but it does not prove the functional equations of the present $n = 3, 4$ packets.

The paper is organized as follows. Section 2 states the source and main theorem. Section 3 proves the point-count and rank identities. Section 4 extracts the leading cohomological logarithm. Section 5 proves the weight–clock bifurcation and twist invariance. Section 6 gives the rank obstruction and the surviving odd skeleton. Section 7 computes the fivefold Hodge ledger, and Section 8 records the Route-A status and claim boundary.

2 The source and the main theorem

Let

$$K = \mathbf{Q}(\rho), \quad \rho^2 + \rho + 1 = 0.$$

For $p > 3$, $p \equiv 1 \pmod{3}$, choose one reduction of ρ in $\mathbf{F}_p$. For $n \in \{2, 3, 4\}$, the source is the ordered $2n$ -variable phase

$$\Phi_{p,n}(x_0, \dots, x_{2n-1}) = 2 \sum_{i=0}^{2n-1} x_i^3 + \sum_{i=0}^{2n-2} x_i x_{i+1} + \rho x_{2n-1} x_0. \quad (1)$$

The last term is the chronological closing edge. We do not average it or replace it by a cyclic transition matrix.

Set

$$\begin{aligned} Z_{p,n} &= \#\Phi_{p,n}^{-1}(0), & C_{p,n} &= 2p^{-(n-1)}Z_{p,n} - 2p^n, \\ d_p &= \frac{p-1}{2}, & c_{p,n} &= \frac{C_{p,n}}{d_p}. \end{aligned} \quad (2)$$

In $\mathbf{P}^{2n-1}$, write

$$\begin{aligned} \mathcal{C}_n &= \sum_{i=0}^{2n-1} x_i^3, & \mathcal{Q}_{n,\rho} &= \sum_{i=0}^{2n-2} x_i x_{i+1} + \rho x_{2n-1} x_0, \\ S_n &= V(\mathcal{C}_n), & Q_n &= V(\mathcal{Q}_{n,\rho}), & X_n &= S_n \cap Q_n. \end{aligned}$$

The predecessor source calculations prove the required smoothness for $n = 2, 3, 4$ outside finite bad sets. Whenever we state a family formula for arbitrary n , smoothness of the displayed member is an explicit hypothesis.

We use geometric Frobenius and the convention that it acts by p on $\mathbf{Q}_\ell(-1)$. Define

$$E_n = \mathbf{Q}_\ell(0) \oplus H_{\text{prim}}^{2n-2}(S_{n,\overline{K}}, \mathbf{Q}_\ell)(n-1), \quad O_n = H^{2n-3}(X_{n,\overline{K}}, \mathbf{Q}_\ell)(n-2). \quad (3)$$

Fix S once and for all as the finite set of rational primes that ramify in K or occur in the inherited source-defined bad-reduction sets for $n = 2, 3, 4$.

Theorem 2.1 (Weight-clock bifurcation). *At every good split prime and for $n = 2, 3, 4$, the following statements hold.*

1. The packets E_n and O_n are pure of weights 0 and 1, respectively, and

$$C_{p,n} = -2\{\text{Tr}(F_p | E_n) + \text{Tr}(F_p | O_n)\}. \quad (4)$$

2. For any smooth member of the displayed family,

$$\text{rank } E_n = \frac{4^n + 5}{3}, \quad \text{rank } O_n = \frac{2(4^n - 4)}{3}, \quad \text{rank}(E_n \oplus O_n) = 4^n - 1.$$

3. Put

$$\ell_n(s) = \sum_{p \equiv 1(3)} c_{p,n} p^{-ns}, \quad F_n(s) = \exp(-\ell_n(s)/n).$$

There is a canonical holomorphic nonzero $H_{n,S}$ on $\text{Re } s > 0$ such that, initially for $\text{Re } s > 1/(2n)$,

$$F_n(s) = \exp\left(\frac{2}{n} \text{Log}_0 L_K^{(S)}(E_n \oplus O_n, ns + 1)\right) H_{n,S}(s). \quad (5)$$

4. The exact denominator tower is $u_{n,j} = ns + j$, $j \geq 1$. A pure weight- w standard reflection maps to the center

$$s_{n,j}(w) = \frac{(w+1)/2 - j}{n}.$$

Consequently, the leading odd centers all equal 0; the leading even centers and the higher odd centers are, respectively,

$$\begin{aligned} (s_{2,1}(0), s_{3,1}(0), s_{4,1}(0)) &= \left(-\frac{1}{4}, -\frac{1}{6}, -\frac{1}{8}\right), \\ s_{n,j}(1) &= -\frac{j-1}{n} \quad (j \geq 2). \end{aligned}$$

5. *These centers are invariant under every consistent Tate relabeling. No semisimple finite-rank K -system in the direct source category can realize the factorwise fractional leading roots at $n = 3, 4$ while both retaining the E_n/O_n weight decomposition and preserving the split-prime trace.*

The final clause is categorical only within its stated source class. Restriction of scalars and Galois-orbit counterpackets change that class and are not excluded. The inherited normalized-semifinite determinant is also not excluded.

For $n = 2$, the exact predecessor factorization is

$$F_2(s) = \zeta_K(2s+1)^7 L(H^1(C/K), 2s+1) H_2(s). \quad (6)$$

The elliptic factors of $H^1(C)$ are modular over $K = \mathbf{Q}(\sqrt{-3})$ by [3, Theorem 1.1]; their standard automorphic analytic package follows from [8, Global Theory, pp. 136–184]. These are the complete $n = 2$ factors. On the initial Euler domain, passing between them and the incomplete factors in (5) changes the residual only by the fixed standard local units, which are holomorphic and nonzero there. No corresponding theorem is invoked for O_3 or O_4 .

Remark 2.2 (Fermat refinement at $n = 3$). The invariant packet is

$$E_3 = \mathbf{Q}_\ell(0) \oplus H_{\text{prim}}^4(S_3)(2).$$

Over K , the primitive term has twenty normalized Tate lines and one rank-two non-Tate Jacobi packet. The extra trivial line gives twenty-one Tate lines in E_3 . This is the classical Fermat/Jacobi decomposition [21, 1]; it is not a claim that the entire primitive cohomology is Tate.

3 Projective cancellation, weights, and ranks

3.1 The four radial strata

For a nonzero direction $[x] \in \mathbf{P}^{2n-1}(\mathbf{F}_p)$,

$$\Phi_{p,n}(tx) = t^2 \{2t\mathcal{C}_n(x) + \mathcal{Q}_{n,\rho}(x)\}.$$

The nonzero roots are counted by four disjoint cases:

condition on $[x]$	number of nonzero t
$\mathcal{C}_n \neq 0, \mathcal{Q}_{n,\rho} \neq 0$	1
$\mathcal{C}_n \neq 0, \mathcal{Q}_{n,\rho} = 0$	0
$\mathcal{C}_n = 0, \mathcal{Q}_{n,\rho} \neq 0$	0
$\mathcal{C}_n = \mathcal{Q}_{n,\rho} = 0$	$p - 1$.

It follows that

$$\begin{aligned} Z_{p,n} &= 1 + \#\{\mathbf{P}^{2n-1} \setminus (S_n \cup Q_n)\} + (p-1)\#X_n \\ &= 1 + P_{2n-1} - \#S_n - \#Q_n + p\#X_n, \end{aligned} \quad (7)$$

where $P_m = 1 + p + \cdots + p^m$.

The quadric Q_n is split in the source rows. Write

$$\#S_n = P_{2n-2} + A_{p,n}, \quad \#Q_n = P_{2n-2} + p^{n-1},$$

$$\#X_n = P_{2n-3} - B_{p,n}.$$

Substitution in (7) cancels every projective Tate polynomial:

$$Z_{p,n} = p^{2n-1} - p^{n-1} - A_{p,n} - pB_{p,n}. \quad (8)$$

Therefore

$$C_{p,n} = -2 - \frac{2A_{p,n}}{p^{n-1}} - \frac{2B_{p,n}}{p^{n-2}}. \quad (9)$$

The two degree-one primes above a split p give the same data. Indeed,

$$x_0 = \rho^2 y_{2n-1}, \quad x_i = y_{i-1} \quad (1 \leq i \leq 2n-1)$$

preserves $\mathcal{C}_n$ and sends the ρ -closing quadric to the ρ^2 -closing quadric.

3.2 Purity and the trace identity

Because S_n has even dimension $2n-2$, its primitive middle trace enters its point count with positive sign. Because X_n has odd dimension $2n-3$, its middle trace enters with negative sign. Hence

$$A_{p,n} = \text{Tr}(F_p | H_{\text{prim}}^{2n-2}(S_n)), \quad B_{p,n} = \text{Tr}(F_p | H^{2n-3}(X_n)).$$

After the twists in (3),

$$e_{p,n} = 1 + \frac{A_{p,n}}{p^{n-1}}, \quad o_{p,n} = \frac{B_{p,n}}{p^{n-2}}$$

are the traces of E_n and O_n . Deligne's theorem [4, Theorem 1.6] makes the packets pure of weights zero and one. Equation (9) becomes

$$C_{p,n} = -2(e_{p,n} + o_{p,n}),$$

proving (4). In particular,

$$e_{p,n} = O(1), \quad o_{p,n} = O(p^{1/2}).$$

3.3 A conditional family identity

For a smooth cubic hypersurface in $\mathbf{P}^{2n-1}$, the primitive Hodge pieces are computed by the Jacobian ring

$$R_n = \mathbf{C}[x_0, \dots, x_{2n-1}] / (x_0^2, \dots, x_{2n-1}^2),$$

whose Hilbert series is $(1+t)^{2n}$ [9, §10, especially Theorem 10.8]. The resulting primitive middle Betti number is

$$b_{2n-2}^{\text{prim}}(S_n) = \frac{4^n + 2}{3}. \quad (10)$$

For a smooth $X_n = (2, 3) \subset \mathbf{P}^{2n-1}$, the normal sequence gives

$$c(TX_n) = \frac{(1+H)^{2n}}{(1+2H)(1+3H)}, \quad \deg X_n = 6.$$

Thus

$$\chi(X_n) = 6[H^{2n-3}] \frac{(1+H)^{2n}}{(1+2H)(1+3H)}.$$

Using

$$\frac{1}{(1+2H)(1+3H)} = -\frac{2}{1+2H} + \frac{3}{1+3H},$$

two finite binomial sums give

$$[H^{2n-3}] \frac{(1+H)^{2n}}{(1+2H)(1+3H)} = \frac{3n+1-4^n}{9}.$$

Consequently,

$$\chi(X_n) = \frac{2(3n+1-4^n)}{3}.$$

Weak Lefschetz [10, Exposé XIV, Corollaire 4.6] leaves only the odd middle Betti number undetermined and gives

$$b_{2n-3}(X_n) = (2n-2) - \chi(X_n) = \frac{2(4^n - 4)}{3}. \quad (11)$$

Equations (10)–(11), together with the trivial line in E_n , prove the rank assertions in Theorem 2.1.

n	rank E_n	rank O_n	total
2	7	8	15
3	23	40	63
4	87	168	255

The computation can be replayed symbolically for larger n , but it is not a smoothness proof for the corresponding Hénon source members.

4 The forced logarithmic L -extraction

Let

$$L_K^{(S)}(V, u) = \prod_{\mathfrak{p} \nmid S} \det(1 - F_{\mathfrak{p}} N \mathfrak{p}^{-u} \mid V^{I_{\mathfrak{p}}})^{-1}$$

be the incomplete standard cohomological Euler product for the frozen set S . An omitted standard factor may be restored only if it is holomorphic and nonzero on the stated domain; we do not absorb arbitrary rational local factors.

For every good split $p \notin S$, (2) and (4) give

$$c_{p,n} = -\frac{4(e_{p,n} + o_{p,n})}{p-1}. \quad (12)$$

Let

$$\ell_n^{(S)}(s) = \sum_{\substack{p \equiv 1(3) \\ p \notin S}} c_{p,n} p^{-ns}.$$

The finite source correction

$$B_{n,S}(s) = -\frac{\ell_n(s) - \ell_n^{(S)}(s)}{n} = -\frac{1}{n} \sum_{\substack{p \in S \\ p \equiv 1(3)}} c_{p,n} p^{-ns} \quad (13)$$

is an entire Dirichlet polynomial. For the good-prime sum,

$$-\frac{\ell_n^{(S)}(s)}{n} = \frac{4}{n} \sum_{\substack{p \equiv 1(3) \\ p \notin S}} (e_{p,n} + o_{p,n}) \sum_{j \geq 1} p^{-ns-j}. \quad (14)$$

At a good split prime $p \notin S$, the two degree-one primes of K have equal traces. The first Euler-logarithm term of

$$\frac{2}{n} \text{Log}_0 L_K^{(S)}(E_n \oplus O_n, ns+1)$$

is hence

$$\frac{4}{n} (e_{p,n} + o_{p,n}) p^{-ns-1},$$

exactly the $j=1$ term of (14).

Proposition 4.1. *The difference*

$$R_n(s) = -\frac{\ell_n(s)}{n} - \frac{2}{n} \text{Log}_0 L_K^{(S)}(E_n \oplus O_n, ns + 1)$$

extends holomorphically to $\text{Re } s > 0$. Thus $H_{n,S}(s) = \exp(R_n(s))$ is holomorphic and nonzero there.

Proof. For $j \geq 2$, the weight-one part left in (14) is

$$O(p^{-n \text{Re } s - 3/2}),$$

and the weight-zero part is smaller. For a split-prime Euler-logarithm power $m \geq 2$, the worst weight-one term is

$$O(p^{-mn \text{Re } s - m/2}).$$

The $m = 2$ prime sum is normally convergent for $\text{Re } s > 0$, and higher m are better. At an inert rational prime outside S , the norm is p^2 , so even its degree-one term satisfies the same boundary. Normal convergence proves the first assertion, and an exponential has no zeros. The remaining term $B_{n,S}$ in (13) is entire, and its exponential is also nonzero. $\square$

The original term in ℓ_n is

$$O(p^{-n \text{Re } s - 1/2}),$$

so ℓ_n converges absolutely for $\text{Re } s > 1/(2n)$. Exponentiating the identity in Proposition 4.1 proves (5).

Remark 4.2 (Meaning of the fractional powers). For $n = 3, 4$, the notation suggested by (5),

$$L_K^{(S)}(E_n \oplus O_n, ns + 1)^{2/n},$$

means only $\exp((2/n) \text{Log}_0 L_K^{(S)})$ on the nonvanishing Euler domain. A local divisor need not be divisible by the denominator, so no single-valued meromorphic root or ordinary rational determinant is inferred.

At $n = 2$, the exponent is integral and $E_2 \simeq \mathbf{Q}_\ell(0)^7$, while $O_2 = H^1(C)$. This gives the exact factorization (6). It will provide a theorem-level witness for the center obstruction in the next section.

5 The weight-clock center tower

For a pure weight- w motive, the standard completed reflection is expected to relate u and $w + 1 - u$ [18, §1.3]. Deligne's conventions place the completed functional equation and Tate shifts in equations (1.2.3) and (3.1.2), respectively [5]. We use those conventions only to compute the *forced center*. Except at $n = 2$, we do not promote the expected reflection to a theorem.

5.1 The canonical index begins at one

Equation (12) contains the exact identity

$$\frac{1}{p-1} = p^{-1} + p^{-2} + p^{-3} + \cdots.$$

Thus the canonical index is

$$j \geq 1, \quad u_{n,j} = ns + j. \tag{15}$$

If $ns' + j = w + 1 - (ns + j)$, then

$$s' = \frac{w + 1 - 2j}{n} - s.$$

The center of this reflection is

$$s_{n,j}(w) = \frac{(w + 1)/2 - j}{n}. \quad (16)$$

For $j = 1$, the three rows are:

weight	$n = 2$	$n = 3$	$n = 4$
$w = 0$	$-1/4$	$-1/6$	$-1/8$
$w = 1$	0	0	0

For $w = 1$ and $j \geq 2$,

$$s_{n,j}(1) = -\frac{j - 1}{n}.$$

Therefore the leading odd rail aligns, the leading even rail bifurcates, and the full tower has no common factorwise standard pure-motive center.

5.2 An unconditional two-factor witness

The factorization (6) contains two completed factors whose analytic equations are known. The Dedekind zeta factor has weight zero and standard center $u = 1/2$; under $u = 2s + 1$, its center is

$$s = \frac{1/2 - 1}{2} = -\frac{1}{4}.$$

The curve H^1 factor has weight one and standard center $u = 1$; under the same clock its center is $s = 0$. Thus a factorwise mismatch is already a theorem at $n = 2$, independent of any conjectural O_3 or O_4 functional equation.

Remark 5.1 (Zero is not the obstruction). The completed Riemann function satisfies $\xi(s) = \xi(1 - s)$; writing $\Xi(z) = \xi(1/2 + z)$ gives a reflection about $z = 0$ [13, §§25.4 and 25.10]. A zero-centered variable is therefore entirely compatible with an RH formulation. The obstruction here is the simultaneous presence of distinct factorwise centers under the same locked clock.

5.3 Tate and half-shift invariance

Let $V(k)$ be an integral Tate twist. It has weight $w - 2k$. To preserve the same local source coefficient, the variable intercept changes from j to $j - k$, because $L(V(k), u) = L(V, u + k)$. Hence

$$\frac{(w - 2k + 1)/2 - (j - k)}{n} = \frac{(w + 1)/2 - j}{n}. \quad (17)$$

The center is invariant.

Automorphic unitary normalization and C -group conventions legitimately use half-sum shifts [2, 16]. Formally setting $k = 1/2$ in (17) still leaves the center fixed. If one changes the weight but holds u fixed, however, the eigenvalue is multiplied by $p^{\mp 1/2}$. That operation changes the frozen Hénon coefficient and is not a repair of the same object.

Corollary 5.2. *No combination of consistent Tate relabelings gives the complete E_n/O_n tower a common factorwise standard pure-motive center.*

The corollary does not exclude a nonfactorwise identity in which additional source-derived factors cancel or reorganize the separate completions.

6 Direct compatible systems and the odd survivor

6.1 A scoped finite-rank obstruction

The leading logarithmic factor in (5) occurs with exponent $2/n$. Suppose it were the standard L -function of a semisimple direct source-native K -compatible system W_n that preserved the same split-prime trace and retained the weight decomposition $E_n \oplus O_n$. Degree-one primes have density one in K . Equality of the leading traces there, Chebotarev density, and Brauer–Nesbitt give [19]

$$n[W_n] = 2[E_n \oplus O_n]$$

in the semisimple representation ring. Purity separates the two weights, so the weightwise dimensions of W_n would have to be

$$\frac{2}{n} \text{rank } E_n, \quad \frac{2}{n} \text{rank } O_n.$$

At $n = 3$, these are

$$\frac{2}{3}(23, 40) = \left(\frac{46}{3}, \frac{80}{3}\right),$$

and at $n = 4$, even the total dimension would be

$$\frac{1}{2}(87 + 168) = \frac{255}{2}.$$

Since the rank of an ordinary compatible system is an integer, the proposed direct K -packets do not exist.

Remark 6.1 (Restriction of scalars). This is not a universal compatible-system no-go. Restriction from K to $\mathbf{Q}$ doubles ranks and changes the organization of primes and local traces. After that change the bare $n = 4$ parity obstruction disappears: the half-multiplicities become 87 and 168. At $n = 3$, the corresponding dimensions $92/3$ and $160/3$ remain nonintegral. Galois-orbit counterpackets, nonfactorwise identities, and the inherited normalized-semifinite determinant are outside the theorem.

6.2 Denominator clearing

The leading odd exponents for $n = 2, 3, 4$ are

$$1, \quad \frac{2}{3}, \quad \frac{1}{2}.$$

Multiplication by their least common denominator 6 gives the integer exponents 6, 4, 3. Let $\Lambda(O_n, u)$ denote a standard completed weight-one factor whenever such a completion is known. Define the formal expected-completion skeleton

$$\mathcal{O}_6(s) = \Lambda(O_2, 2s + 1)^6 \Lambda(O_3, 3s + 1)^4 \Lambda(O_4, 4s + 1)^3. \quad (18)$$

If all three factors satisfy their expected standard reflections

$$\Lambda(O_n, u) = \varepsilon_n \Lambda(O_n, 2 - u),$$

then

$$\mathcal{O}_6(s) = \varepsilon \mathcal{O}_6(-s).$$

This implication is exact, but its premise is proved here only for O_2 . Equation (18) is therefore a denominator-cleared formal skeleton, not a new Hasse–Weil theorem.

Product stability is natural in motivic λ -ring settings [11, Proposition 6.1], but it presupposes factors that already possess the relevant rationality and functional equations. It does not construct the missing O_3, O_4 analytic packages.

6.3 Why a formal Gamma product is insufficient

Serre's Hodge-theoretic Gamma ledger and Deligne's expected completion specify which finite products would accompany an individual pure motive [18, 5]. The complete Hénon denominator tower, however, produces an infinite list of shifted centers. Turning that list into a regularized product requires a spectral zeta function with a meromorphic continuation at the regularization point and fixed branch and multiplicity data. These are part of the definition of a zeta-regularized product [14].

Deninger expressed motivic Gamma and local factors through regularized determinants [6, 7]. This is an important structural precedent, but it does not license a determinant for an unconstructed Hénon archimedean spectrum. We therefore stop at the finite odd skeleton (18).

7 The fivefold Hodge ledger and the next gate

The primitive cubic Hodge pieces follow from the Jacobian ring, while the odd complete-intersection pieces follow from weak Lefschetz and Hirzebruch–Riemann–Roch. After the twists in (3), the three rows are:

$$\begin{array}{l|l} E_2 & (0, 0)^7 \\ O_2 & (1, 0)^4 + (0, 1)^4 \\ E_3 & (1, -1)^1 + (0, 0)^{21} + (-1, 1)^1 \\ O_3 & (1, 0)^{20} + (0, 1)^{20} \\ E_4 & (1, -1)^8 + (0, 0)^{71} + (-1, 1)^8 \\ O_4 & (2, -1)^1 + (1, 0)^{83} + (0, 1)^{83} + (-1, 2)^1. \end{array}$$

At the unique complex place of K , the corresponding expected Deligne–Serre Gamma ledger in the present motivic normalization is

$$\Gamma_{\mathbf{C}}(u) = 2(2\pi)^{-u}\Gamma(u),$$

and

$$\begin{array}{l|l} E_2 & \Gamma_{\mathbf{C}}(u)^7 \\ O_2 & \Gamma_{\mathbf{C}}(u)^8 \\ E_3 & \Gamma_{\mathbf{C}}(u)^{21}\Gamma_{\mathbf{C}}(u+1)^2 \\ O_3 & \Gamma_{\mathbf{C}}(u)^{40} \\ E_4 & \Gamma_{\mathbf{C}}(u)^{71}\Gamma_{\mathbf{C}}(u+1)^{16} \\ O_4 & \Gamma_{\mathbf{C}}(u)^{166}\Gamma_{\mathbf{C}}(u+1)^2. \end{array}$$

This is an expected sector Gamma ledger obtained from the Hodge types [18, 5]. It neither proves the $n = 3, 4$ functional equations nor defines an infinite Gamma product for the full Hénon denominator tower.

We give a finite independent derivation of the last line.

7.1 The χ_y calculation

For $X_4 = (2, 3) \subset \mathbf{P}^7$, define the Hirzebruch characteristic series

$$Q_y(x) = \frac{x(1 + ye^{-x})}{1 - e^{-x}}.$$

Hirzebruch–Riemann–Roch [12] gives

$$\chi_y(X_4) = 6[H^5] \frac{Q_y(H)^8}{(1+y)Q_y(2H)Q_y(3H)}. \quad (19)$$

Only terms through degree five contribute. The finite expansion

$$Q_y(x) = (1 + y) + \frac{1 - y}{2}x + \frac{1 + y}{12}x^2 - \frac{1 + y}{720}x^4 + O(x^6)$$

in (19) yields

$$\chi_y(X_4) = 1 - 82y^2 + 82y^3 - y^5. \quad (20)$$

Weak Lefschetz and Hodge symmetry leave two middle Hodge numbers

$$a = h^{4,1} = h^{1,4}, \quad b = h^{3,2} = h^{2,3}.$$

In terms of a, b , the same genus is

$$1 + (a - 1)y + (1 - b)y^2 + (b - 1)y^3 + (1 - a)y^4 - y^5.$$

Comparing with (20) gives

$$a = 1, \quad b = 83.$$

Twisting $H^5(X_4)$ by 2 proves the displayed O_4 row.

7.2 The C52 projector gate

The types $(2, -1)$ and $(-1, 2)$ show that O_4 is not itself the H^1 of an abelian variety. Its Hodge ledger nevertheless suggests a rank-two extreme part and a rank-166 level-one part:

$$O_4^{\text{ext}} : (2, -1) + (-1, 2), \quad O_4^{\text{lev1}} : (1, 0)^{83} + (0, 1)^{83}.$$

A Hodge decomposition does not imply an algebraic splitting over K . Nor does a numerical projector at one prime define a compatible system.

Proposition 7.1 (C52 gate). *The next positive route requires a K -rational algebraic correspondence whose induced projectors on all ℓ -adic realizations are compatible and separate O_4^{ext} from O_4^{lev1} . Absent such a construction, the appropriate negative target is a monodromy or endomorphism theorem obstructing that algebraic splitting.*

Proposition 7.1 is a research gate, not an existence theorem. It is large enough to change the analytic route: a rank-two extreme factor could be tested for a Hecke or automorphic realization, while the level-one factor could be compared with an abelian-type motive. Additional unstructured prime counting would not decide algebraicity.

8 Route-A evaluation and Hilbert–Pólya scope

C51 changes the structural evaluation, not the analytic half-plane. The normalized Euler object remains the one continued in the predecessor to $\text{Re } s > 1/5$. Its operator realization remains a tenth-order graded regularized determinant relative to the normalized faithful semifinite trace. The regularized-determinant background is standard [20, Chapter 9]; the source-specific Schatten thresholds and counterterms are inherited rather than reproved here.

Gate	Verdict	C51 evidence and boundary
A1	weak	The four-, six-, and eight-step source chronology is exact, but split primes are arithmetic fibres rather than primitive orbits of one real map.
A2	analytic determinant	The C50 continuation to $\text{Re } s > 1/5$ and $\text{Det}_{10,\tau,\text{gr}}$ are inherited; fractional cohomological powers are only Log_0 -germs.
A3	partial analytic structure	Leading odd centers align and the $n = 2$ component equations are proved; the even rail and higher tower do not align, and the full equation is open.
A4	natural quantization	The unitary finite-place blocks, prime clock, and normalized trace are source-native; no self-adjoint global generator is constructed.

The exact evaluator tuple is

A1_WEAK, A2_ANALYTIC_DETERMINANT,
A3_PARTIAL_ANALYTIC_STRUCTURE,
A4_NATURAL_QUANTIZATION

with overall status ROUTE_A_EXPLORATORY.

The A3 delta can be summarized without changing the enum:

leading odd $j = 1$ center alignment	proved,
full factorwise standard center	refuted,
full denominator-tower alignment	refuted,
direct K -packet repair at $n = 3, 4$	refuted in scope,
$n = 2$ extracted factor equation	proved,
$n = 3, 4$ odd equations	open,
full Hénon functional equation	open.

The paper proves no new zero-free region, no continuation through $\text{Re } s = 1/5$, no full Hénon Gamma factor, no Riemann divisor, and no Riemann–von Mangoldt law. The normalized-semifinite determinant is not a classical Fredholm determinant on the claimed half-plane. Route B is not authorized.

The next large door is Proposition 7.1. It is deliberately sharper than a request to compute more primes: construct a K -rational algebraic projector with ℓ -compatible realizations, or prove a monodromy or endomorphism obstruction to it.

Declarations and limitations

Data and code availability

The source equations, theorem and derivation packages, exact producer, independent checker, mutation tests, Route-A records, and compiled manuscript are released together in the repository directory *henon_dynamics/henon_mu3_weight_clock_bifurcation*. No proprietary data, Riemann-zero table, or floating-point fit is used. Machine certificates replay finite symbolic identities and schema semantics; they do not replace the cited purity, Lefschetz, modularity, or Riemann–Roch theorems.

Ethics statement

This mathematical study involved no human participants, personal data, animals, clinical intervention, or biological material. Ethics approval and informed consent were not applicable.

Author contributions (CRediT)

The Hénnon Zeta Research Program performed conceptualization, methodology, formal analysis, software, validation, investigation, data curation, and writing of the original draft and revisions. Claim promotion followed independent mathematical red-team, source-audit, producer/checker, and mutation-testing procedures.

Funding

No external funding was reported for this study.

Conflict of interest

The authoring research program reports no financial or nonfinancial conflict of interest.

AI-use statement

AI systems were used for mathematical exploration, exact-code generation, primary-source triage, adversarial review, and manuscript drafting. AI output was not accepted as proof by itself. Theorem-level computational claims are backed by exact arithmetic and replayable certificates, and external mathematical inputs are assigned to cited primary sources. No AI system is listed as a human author.

Limitations

The construction has six principal limitations. First, split primes index arithmetic fibres, not primitive cycles of one classical Hénnon map. Second, the common-center result applies only to the leading odd rail; the even rail and later denominator terms bifurcate. Third, the $n = 3, 4$ odd functional equations are expected rather than proved. Fourth, the finite-rank obstruction applies only to semisimple direct source-native K -packets retaining the two-weight decomposition and trace. Restriction of scalars, Galois counterpackets, and nonfactorwise identities remain open. Fifth, the inherited continuation may have zeros and stops at $\operatorname{Re} s > 1/5$. Sixth, no full archimedean completion, self-adjoint operator, Riemann divisor, or zero-counting law is constructed.

A Finite exact replay formulas

This appendix records the finite algebra required by the release checker. It does not replace the external geometric theorems cited in the text.

A.1 Rank controls

For a smooth family member, the exact rank formulas are

n	$b^{\text{prim}}(S_n)$	$b^{\text{mid}}(X_n)$	$\operatorname{rank}(E_n \oplus O_n)$
2	6	8	15
3	22	40	63
4	86	168	255.

The general identities

$$b^{\text{prim}}(S_n) = \frac{4^n + 2}{3}, \quad b^{\text{mid}}(X_n) = \frac{2(4^n - 4)}{3}$$

may be checked by finite polynomial arithmetic. Symbolic controls for $n > 4$ certify the formula conditional on smoothness; they do not certify source smoothness.

A.2 Center controls

With $j \geq 1$,

$$s_{n,j}(w) = \frac{(w+1)/2 - j}{n}.$$

The leading center spectrum is exactly

$$\left\{-\frac{1}{4}, -\frac{1}{6}, -\frac{1}{8}, 0\right\}.$$

The mutation $j = 0$ changes the source denominator expansion and must be rejected. The consistent twist mutation

$$(w, j) \mapsto (w - 2k, j - k)$$

must leave every center unchanged, while a fixed-clock half-weight mutation must change the local coefficient and be rejected.

A.3 Hodge control

Truncate

$$Q_y(x) = (1 + y) + \frac{1 - y}{2}x + \frac{1 + y}{12}x^2 - \frac{1 + y}{720}x^4 + O(x^6)$$

at degree five and evaluate

$$6[H^5] \frac{Q_y(H)^8}{(1 + y)Q_y(2H)Q_y(3H)}.$$

The exact output is

$$1 - 82y^2 + 82y^3 - y^5.$$

Comparing with the weak-Lefschetz template recovers

$$h^{4,1} = 1, \quad h^{3,2} = 83.$$

A.4 Checker scope

The independent checker is required to fail closed on altered trace signs, rank formulas, tower indexing, center values, Tate shifts, fractional-root scope, Hodge coefficients, unknown schema keys, and overclaim fields. It does not rederive Deligne purity, weak Lefschetz, the modularity theorem, or an $n = 3, 4$ Hasse–Weil functional equation.

References

- [1] Lars Brünjes. On the zeta function of forms of Fermat equations. arXiv:math/0301186, 2003.
- [2] Kevin Buzzard and Toby Gee. The conjectural connections between automorphic representations and Galois representations. arXiv:1009.0785, 2010.
- [3] Ana Caraiani and James Newton. On the modularity of elliptic curves over imaginary quadratic fields. arXiv:2301.10509, 2025. Version 3, Theorem 1.1.
- [4] Pierre Deligne. La conjecture de Weil. I. *Publications Mathématiques de l’IHÉS*, 43:273–308, 1974.
- [5] Pierre Deligne. Valeurs de fonctions L et périodes d’intégrales. In *Automorphic Forms, Representations and L-Functions*, volume 33 of *Proceedings of Symposia in Pure Mathematics*, pages 313–346. American Mathematical Society, 1979.

- [6] Christopher Deninger. On the Γ -factors attached to motives. *Inventiones Mathematicae*, 104(2):245–261, 1991.
- [7] Christopher Deninger. Local L-factors of motives and regularized determinants. *Inventiones Mathematicae*, 107(1):135–150, 1992.
- [8] Roger Godement and Hervé Jacquet. *Zeta Functions of Simple Algebras*, volume 260 of *Lecture Notes in Mathematics*. Springer, Berlin, 1972.
- [9] Phillip A. Griffiths. On the periods of certain rational integrals. II. *Annals of Mathematics*, 90(3):496–541, 1969.
- [10] Alexandre Grothendieck and Michèle Raynaud. *SGA 2: Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux*. North-Holland, Amsterdam, 1968. Exposé XIV.
- [11] Franziska Heinloth. A note on functional equations for zeta functions with values in Chow motives. *Annales de l’Institut Fourier*, 57(6):1927–1945, 2007.
- [12] Friedrich Hirzebruch. *Topological Methods in Algebraic Geometry*. Springer, Berlin, 3rd, second corrected printing edition, 1995.
- [13] NIST Digital Library of Mathematical Functions. Chapter 25: Zeta and related functions. <https://dlmf.nist.gov/25>. Sections 25.4 and 25.10, accessed 14 August 2026.
- [14] J. R. Quine, S. H. Heydari, and R. Y. Song. Zeta regularized products. *Transactions of the American Mathematical Society*, 338(1):213–231, 1993.
- [15] David Ruelle. Zeta-functions for expanding maps and Anosov flows. *Inventiones Mathematicae*, 34:231–242, 1976.
- [16] Peter Sarnak, Sug Woo Shin, and Nicolas Templier. Families of L-functions and their symmetry. arXiv:1401.5507, 2014.
- [17] Atle Selberg. Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces with applications to Dirichlet series. *Journal of the Indian Mathematical Society*, 20:47–87, 1956.
- [18] Jean-Pierre Serre. Facteurs locaux des fonctions zêta des variétés algébriques (définitions et conjectures). *Séminaire Delange–Pisot–Poitou*, 11(2):1–15, 1969–1970. Exposé 19.
- [19] Jean-Pierre Serre. Quelques applications du théorème de densité de Chebotarev. *Publications Mathématiques de l’IHÉS*, 54:123–201, 1981.
- [20] Barry Simon. *Trace Ideals and Their Applications*, volume 120 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2 edition, 2005.
- [21] André Weil. Jacobi sums as Grössencharaktere. *Transactions of the American Mathematical Society*, 73(3):487–495, 1952.